

✧ **De-Morgan's law :**

(a) $(A \cup B)' = A' \cap B'$

(b) $(A \cap B)' = A' \cup B'$

Proof : (a) Let x be any arbitrary element of the set $(A \cup B)'$ then,

$$x \in (A \cup B)' \Rightarrow \{(x : x \notin (A \cup B))\}$$

$$\Rightarrow x \notin A \text{ and } x \notin B$$

$$\Rightarrow x \in A' \text{ and } x \in B'$$

$$\Rightarrow x \in A' \cap B'$$

$$\therefore (A \cup B)' \subset A' \cap B' \quad \dots(i)$$

Again let y be any arbitrary element of the set $A' \cap B'$ then,

$$y \in A' \cap B' \Rightarrow y \in A' \text{ and } y \in B'$$

$$\Rightarrow y \notin A \text{ and } y \notin B \Rightarrow y \notin (A \cup B)$$

$$\Rightarrow y \in (A \cup B)'$$

$$\therefore (A' \cap B') \subseteq (A \cup B)' \quad \dots(ii)$$

From (i) and (ii), we have,

$$(A \cup B)' = A' \cap B'.$$

(b) Let x be any arbitrary element of the set $(A \cap B)'$ then,

$$x \in (A \cap B)' \Rightarrow x \notin (A \cap B)$$

$$\Rightarrow x \notin A \text{ or } x \notin B$$

$$\Rightarrow x \in A' \text{ or } x \in B'$$

$$\Rightarrow x \in (A' \cup B')$$

$$\therefore (A \cap B)' \subseteq (A' \cup B')$$

Again let y be arbitrary element of the set $A' \cup B'$ then,

$$y \in A' \cup B' \Rightarrow x \in A' \text{ or } x \in B'$$

$$\Rightarrow x \notin A \text{ or } x \notin B$$

$$\Rightarrow x \text{ does not belong to both } A \text{ and } B$$

$$\Rightarrow x \notin (A \cap B)$$

$$\Rightarrow x \in (A \cap B)'$$

$$\therefore A' \cup B' \subseteq (A \cap B)'$$

From (i) and (ii), we have,

$$(A \cap B)' = A' \cup B'.$$

(2) If A, B, C are three sets, then,

$$(a) \quad A - (B \cup C) = (A - B) \cap (A - C)$$

$$(b) \quad A - (B \cap C) = (A - B) \cup (A - C).$$

Proof : (a) Let x be any arbitrary element of the set $A - (B \cup C)$, then

$$x \in A - (B \cup C) \Rightarrow x \in A \text{ and } x \notin B \cup C$$

$$\Rightarrow x \in A \text{ and } (x \notin B \text{ and } x \notin C)$$

$$\Rightarrow (x \in A \text{ and } x \notin B) \text{ and } (x \in A \text{ and } x \notin C)$$

$$\Rightarrow x \in (A - B) \text{ and } x \in (A - C)$$

$$\Rightarrow x \in [(A - B) \cap (A - C)]$$

$$\therefore A - (B \cup C) \subseteq (A - B) \cap (A - C) \quad \dots(i)$$

Again let y be any arbitrary element of the set $(A - B) \cap (A - C)$. Then,

$$y \in (A - B) \cap (A - C) \Rightarrow y \in (A - B) \text{ and } y \in (A - C)$$

$$\Rightarrow (y \in A \text{ and } y \notin B) \text{ and } (y \in A \text{ and } y \notin C)$$

$$\Rightarrow y \in A \text{ and } (y \notin B \text{ and } y \notin C)$$

$$\Rightarrow y \in A \text{ and } y \notin (B \cup C)$$

$$\Rightarrow y \in A - (B \cup C)$$

$$\therefore (A - B) \cap (A - C) \subseteq A - (B \cup C) \quad \dots(ii)$$

From (i) and (ii) we have,

$$A - (B \cup C) = (A - B) \cap (A - C).$$

(b) Let x be any arbitrary element of $A - (B \cap C)$, then

$$x \in A - (B \cap C) \Rightarrow x \in A \text{ and } x \notin (B \cap C)$$

$$\Rightarrow x \in A \text{ and } (x \notin B \text{ or } x \notin C)$$

$$\Rightarrow$$

$$(x \in A \text{ and } x \notin B) \text{ or } (x \in A \text{ and } x \notin C)$$

$$\Rightarrow x \in (A - B) \text{ or } x \in (A - C)$$

$$\Rightarrow x \in (A - B) \cup (A - C)$$

$$\therefore A - (B \cap C) \subseteq (A - B) \cup (A - C).$$

Again let y be any arbitrary element of $(A - B) \cup (A - C)$, then,

$$y \in (A - B) \cup (A - C) \Rightarrow [y \in (A - B)] \text{ or } [y \in (A - C)]$$

$$\Rightarrow (y \in A \text{ and } y \notin B) \text{ or } (y \in A \text{ and } y \notin C)$$

$$\Rightarrow y \in A \text{ and } (y \notin B \text{ or } y \notin C)$$

$$\Rightarrow y \in A \text{ and } y \notin (B \cap C)$$

$$\therefore (A - B) \cup (A - C) \subseteq A - (B \cap C) \text{(ii)}$$

From (i) and (ii), we have

$$A - (B \cap C) = (A - B) \cup (A - C).$$

(3) Prove that

$$A - (B - C) = (A - B) \cup (A \cap C).$$

Proof : Let x be any arbitrary element of the set $A - (B - C)$, then,

$$\begin{aligned} x \in A - (B - C) &\Rightarrow x \in A \text{ and } x \notin (B - C) \\ &\Rightarrow x \in A \text{ and } x \notin (B \cap C') \\ &\Rightarrow x \in A \text{ and } (x \notin B \text{ or } x \notin C') \\ &\Rightarrow (x \in A \text{ and } x \notin B) \text{ or } \\ &\quad (x \in A \text{ and } x \notin C') \\ &\Rightarrow (x \in A \text{ and } x \notin B) \text{ or } \\ &\quad (x \in A \text{ and } x \in C) \\ &\Rightarrow x \in (A - B) \text{ or } x \in (A \cap C) \\ &\Rightarrow x \in (A - B) \cup (A \cap C) \end{aligned}$$

$$\therefore A - (B - C) \subseteq (A - B) \cup (A \cap C)$$

Again let y be any arbitrary element of the set $(A - B) \cup (A \cap C)$, then,

$$\begin{aligned} y \in (A - B) \cup (A \cap C) &\Rightarrow y \in (A - B) \text{ or } y \in (A \cap C) \\ &\Rightarrow (y \in A \text{ and } y \notin B) \text{ or } (y \in A \text{ and } y \in C) \\ &\Rightarrow (y \in A \text{ and } y \notin B) \text{ or } (y \in A \text{ and } y \notin C') \\ &\Rightarrow y \in A \text{ and } (y \notin B \text{ or } y \notin C') \\ &\Rightarrow y \in A \text{ and } y \notin (B \cap C') \\ &\Rightarrow y \in A \text{ and } y \notin (B - C) \\ &\Rightarrow y \in A - (B - C) \end{aligned}$$

$$\therefore (A - B) \cup (A \cap C) \subseteq A - (B - C)$$

From (i) and (ii) we have,

$$A - (B - C) = (A - B) \cup (A \cap C).$$

Ex. 1 : For any two sets A and B, prove that

$$A \Delta B = (A \cup B) - (A \cap B) \text{ or}$$

$$(A - B) \cup (B - A) = (A \cup B) - (A \cap B).$$

Sol. : Let x be any arbitrary element of the set

$$(A - B) \cup (B - A). \text{ Then, } x \in (A - B) \cup (B - A)$$

$$x \in (A - B) \text{ or } x \in (B - A)$$

$$\Rightarrow (x \in A \text{ and } x \notin B) \text{ or } (x \in B \text{ and } x \notin A)$$

$$\Rightarrow (x \in A \text{ or } x \in B) \text{ and } (x \notin A \text{ and } x \notin B)$$

$$\Rightarrow (x \in A \cup B) \text{ and } x \notin (A \cap B)$$

$$\Rightarrow x \in (A \cup B) - (A \cap B)$$

$$\therefore (A - B) \cup (B - A) \subseteq (A \cup B) - (A \cap B)$$

Again let x be any arbitrary element of $(A \cup B) - (A \cap B)$.

Then,

$$x \in (A \cup B) - (A \cap B) \Rightarrow (x \in A \cup B) \text{ and } (x \notin A \cap B)$$

$$\Rightarrow (x \in A \text{ or } x \in B) \text{ and } (x \notin A \text{ and } x \notin B)$$

$$\Rightarrow (x \in A \text{ and } x \notin B) \text{ or } (x \in B \text{ and } x \notin A)$$

$$\Rightarrow x \in (A - B) \text{ or } x \in (B - A)$$

$$x \in (A - B) \cup (B - A)$$

$$\therefore (A \cup B) - (A \cap B) \subseteq (A - B) \cup (B - A)$$

From (i) and (ii), we have,

$$(A - B) \cup (B - A) = (A \cup B) - (A \cap B)$$

$$A \Delta B = (A \cup B) - (A \cap B).$$

Ex. 2 : If A and B are two sets, then prove that

$$A \cup B = (A - B) \cup (B - A) \cup (A \cap B).$$

Sol. : Let x be any arbitrary element of sets $A \cup B$, then,

$$x \in A \cup B \Rightarrow x \in A \text{ or } x \in B$$

$$\Rightarrow (x \in A \text{ and } x \notin B) \text{ or } (x \in B \text{ and } x \notin A)$$

'or'

$$(x \in A) \text{ and } (x \in B)$$

$$\Rightarrow x \in (A - B) \text{ or } x \in (B - A) \text{ or } x \in (A \cap B)$$

$$\Rightarrow x \in [(A - B) \cup (B - A) \cup (A \cap B)]$$

$$\therefore A \cup B \subseteq [(A - B) \cup (B - A) \cup (A \cap B)] \quad \dots(i)$$

Again let y be any arbitrary element of set

$$[(A - B) \cup (B - A) \cup (A \cap B)].$$

$$\text{Then } y \in [(A - B) \cup (B - A) \cup (A \cap B)]$$

$$\Rightarrow y \in (A - B) \text{ or } y \in (B - A) \text{ or } y \in (A \cap B)$$

$$\Rightarrow (y \in A \text{ and } y \notin B) \text{ or } (y \in B \text{ and } y \notin A) \text{ or } (y \in A \text{ and } y \in B)$$

$$\Rightarrow y \in A \text{ or } y \in B \text{ or } y \in A \text{ and } y \in B.$$

$$\Rightarrow y \in (A \cup B) \quad [\text{by def. of union set}]$$

$$\therefore [(A - B) \cup (B - A) \cup (A \cap B)] \subset (A \cup B)$$

From (i) and (ii), we have,

$$A \cup B = (A - B) \cup (B - A) \cup (A \cap B).$$

Ex. 3 : If $A = \{1, 2\}$, $B = \{2, 3\}$, $C = \{3, 4\}$, then evaluate

$$(i) \quad (A \times B) \cup (A \times C)$$

$$(ii) \quad (A \times B) \cap (A \times C).$$

$$\text{Sol. : } A \times B = \{1, 2\} \times \{2, 3\}$$

$$= \{(1, 2), (1, 3), (2, 2), (2, 3)\}$$

$$A \times C = \{1, 2\} \times \{3, 4\}$$

$$= \{(1, 3), (1, 4), (2, 3), (2, 4)\}$$

$$\therefore (i) \quad (A \times B) \cup (A \times C)$$

$$= \{(1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4)\}$$

$$(ii) \quad (A \times B) \cap (A \times C)$$

$$= \{(1, 3), (2, 3)\}.$$

Ex. 4 : Let $A = \{2, 3\}$, $B = \{4, 5\}$, $C = \{5, 6\}$, then evaluate

$$(i) \quad A \times (B \cup C),$$

$$(ii) \quad A \times (B \cap C),$$

$$(iii) \quad (A \times B) \cup (B \times C).$$

$$\text{Sol. : } B \cup C = \{4, 5\} \cup \{5, 6\} = \{4, 5, 6\}$$

$$B \cap C = \{4, 5\} \cap \{5, 6\} = \{5\}$$

$$\begin{aligned} \text{(i) } A \times (B \cup C) &= \{2, 3\} \times \{4, 5, 6\} \\ &= \{(2, 4), (2, 5), (2, 6), (3, 4), (3, 5), (3, 6)\} \end{aligned}$$

$$\begin{aligned} \text{(ii) } A \times (B \cap C) &= \{2, 3\} \times \{5\} \\ &= \{(2, 5), (3, 5)\} \end{aligned}$$

$$\begin{aligned} \text{(iii) } A \times B &= \{2, 3\} \times \{5, 6\} \\ &= \{(2, 5), (2, 6), (3, 5), (3, 6)\} \end{aligned}$$

$$\begin{aligned} B \times C &= \{4, 5\} \times \{5, 6\} \\ &= \{(4, 5), (4, 6), (5, 5), (5, 6)\} \end{aligned}$$

$$\begin{aligned} \therefore (A \times B) \cup (B \times C) &= \{(2, 5), (2, 6), (3, 5), (3, 6), \\ &\quad (4, 5), (4, 6), (5, 5), (5, 6)\}. \end{aligned}$$

Ex. 5 : If A, B, C are any three sets, prove that

$$(A \cup B) \times C = (A \times C) \cup (B \times C).$$

Sol. : Let (x, y) be any arbitrary element of $(A \cup B) \times C$ then $(x, y) \in (A \cup B) \times C$

$$\Rightarrow x \in (A \cup B) \text{ and } y \in C$$

$$\Rightarrow (x \in A \text{ or } x \in B) \text{ and } y \in C$$

$$\Rightarrow (x \in A \text{ And } y \in C) \text{ or } (x \in B \text{ and } y \in C)$$

$$\Rightarrow (x, y) \in (A \times C) \text{ or } (x, y) \in (B \times C)$$

$$\Rightarrow (x, y) \in (A \times C) \cup (B \times C)$$

$$\therefore (A \cup B) \times C \subseteq (A \times C) \cup (B \times C).$$

Again let (z, u) be any arbitrary element of $(A \times C) \cup (B \times C)$. Then $(z, u) \in (A \times C) \cup (B \times C)$.

$$\Rightarrow (z, u) \in (A \times C) \text{ or } (z, u) \in (B \times C)$$

$$\Rightarrow (z \in A \text{ and } u \in C) \text{ or } (z \in B \text{ and } u \in C)$$

$$\Rightarrow (z \in A \text{ or } z \in B) \text{ and } u \in C$$

$$\Rightarrow (z, u) \in (A \cup B) \times C$$

$$\therefore (A \times C) \cup (B \times C) \subseteq (A \cup B) \times C \quad \dots(ii)$$

From (i) And (ii), we have,

$$(A \cup B) \times C = (A \times C) \cup (B \times C).$$

Ex. 6 : If A , B and C be any three sets, then, prove that

$$(A \cap B) \times C = (A \times C) \cap (B \times C).$$

Sol. : Let (x, y) be any element of $(A \cap B) \times C$, then
 $(x, y) \in (A \cap B) \times C$

$$\Rightarrow x \in (A \cap B) \text{ and } y \in C$$

$$\Rightarrow (x \in A \text{ and } x \in B) \text{ and } y \in C$$

$$\Rightarrow (x \in A \text{ and } y \in C) \text{ and } (x \in B \text{ and } y \in C)$$

$$\Rightarrow (x, y) \in (A \times C) \text{ and } (x, y) \in (B \times C)$$

$$\Rightarrow (x, y) \in (A \times C) \cap (B \times C) \quad \dots(i)$$

$$\therefore (A \cap B) \times C \subseteq (A \times C) \cap (B \times C).$$

Again let (z, u) be any arbitrary element of $(A \times C) \cap (B \times C)$.

$$\text{Then } (z, u) \in (A \times C) \cap (B \times C)$$

$$\Rightarrow (z, u) \in (A \times C) \text{ and } (z, u) \in (B \times C)$$

$$\Rightarrow (z \in A \text{ and } u \in C) \text{ and } (z \in B \text{ and } u \in C)$$

$$\Rightarrow (z \in A \text{ and } z \in B) \text{ and } u \in C$$

$$\Rightarrow (z, u) \in (A \cap B) \times C \quad \dots(ii)$$

From (i) and (ii), we have,

$$(A \cap B) \times C = (A \times C) \cap (B \times C).$$

Ex. 7 : Prove that,

$$(A \cap B) \times (C \cap D) = (A \times C) \cap (B \times D).$$

Sol. : Let (x, y) be any arbitrary element of

$$(A \cap B) \times (C \cap D) \text{ then,}$$

$$(x, y) \in (A \cap B) \times (C \cap D)$$

$$\Rightarrow x \in A \cap B, y \in C \cap D \text{ [by def. of product of sets]}$$

$$\Rightarrow (x \in A \text{ and } x \in B), (y \in C \text{ and } y \in D)$$

$$\Rightarrow (x \in A, y \in C) \text{ and } (x \in B, y \in D)$$

$$\Rightarrow (x, y) \in (A \times C) \text{ and } (x, y) \in (B \times D)$$

$$\Rightarrow (x, y) \in (A \times C) \cap (B \times D)$$

$$\therefore (A \cap B) \times (C \cap D) \subseteq (A \times C) \cap (B \times D) \quad \dots(i)$$

Again let (z, u) be any arbitrary element of

$$(A \times C) \cap (B \times D) \text{ then } (z, u) \in (A \times C) \cap (B \times D)$$

$$\Rightarrow (z, u) \in (A \times C) \text{ And } (z, u) \in (B \times D)$$

[By def. of intersection of sets]

$$\Rightarrow (z \in A, u \in C) \text{ and } (z \in B, u \in D)$$

$$\Rightarrow (z \in A \text{ and } z \in B), (u \in C \text{ and } u \in D)$$

$$\Rightarrow z \in (A \cap B), u \in (C \cap D)$$

$$\Rightarrow (z, u) \in (A \cap B) \times (C \cap D)$$

[By def. of product of sets]

$$\therefore (A \times C) \cap (B \times D) \subseteq (A \cap B) \times (C \cap D) \quad \dots(ii)$$

From (i) And (ii), we get,

$$(A \cap B) \times (C \cap D) = (A \times C) \cap (B \times D).$$

Ex. 8 : If A and B be non empty sets, then show that $A \times B = B \times A$ if $A = B$.

Sol. : Let $A \times B = B \times A$ and also let $x \subseteq A$.

$$\text{Then } x \in A \Rightarrow (x, b) \in A \times B \quad \forall b \in B$$

$$\text{But, } A \times B = B \times A$$

$$\Rightarrow (x, b) \in B \times A$$

$$\Rightarrow x \in B, \therefore A \subseteq B. \quad \dots(i)$$

Again let $y \in B$

$$\text{Then, } y \in B \Rightarrow (a, y) \in A \times B, \quad \forall a \in A$$

$$\text{But, } A \times B = B \times A$$

$$\Rightarrow (a, y) \in B \times A$$

$$\Rightarrow y \in A, \quad B \subseteq A \quad \dots(ii)$$

From (i) and (ii), we have,

$$A = B, \text{ if } A \times B = B \times A.$$

Converse, let $A = B$

$$\text{Then } A \times B = A \times A$$

$$B \times A = A \times A$$

$$\therefore A \times B = B \times A$$

Hence, $A \times B = B \times A$ if $A = B$.

Ex. 9 : If A, B, C are three sets, then prove that

$$(A - B) \times C = (A \times C) - (B \times C).$$

Sol. : Let $(x, y) \Rightarrow (A - B) \times C$

$$\Rightarrow x \in (A - B) \text{ and } y \in C$$

$$\Rightarrow (x \in A \text{ and } x \notin B) \text{ and } y \in C$$

$$\Rightarrow (x \in A \text{ and } x \notin B) \text{ and } y \in C$$

$$\Rightarrow (x \in A \text{ and } y \in C) \text{ and } (x \notin B \text{ and } y \in C)$$

$$\Rightarrow (x, y) \in (A \times C) \text{ and } (x, y) \notin (B \times C)$$

$$\Rightarrow (x, y) \in (A \times C) - (B \times C)$$

$$\therefore (A - B) \times C \subseteq (A \times C) - (B \times C) \quad \dots(i)$$

Again let $(x, y) \in (A \times C) - (B \times C)$

$$\Rightarrow (x, y) \in A \times C \text{ and } (x, y) \notin (B \times C)$$

$$\Rightarrow (x \in A \text{ and } y \in C) \text{ and } (x \notin B \text{ and } y \in C)$$

$$\Rightarrow (x \in A \text{ and } x \notin B) \text{ and } y \in C$$

$$\Rightarrow x \in (A - B) \text{ and } y \in C$$

$$\Rightarrow (x, y) \in (A - B) \times C$$

$$\therefore (A \times C) - (B \times C) \subseteq (A - B) \times C \quad \dots(ii)$$

From (i) and (ii), we have,

$$(A - B) \times C = (A \times C) - (B \times C).$$

Ex. 10 : If $A = \{x \in N / x < 6\}$ and $B = \{x \in N / (x + 1)^2 < 49\}$ then prove that $A = B$.

Sol. : $A = \{1, 2, 3, 4, 5\}$

$B = \{1, 2, 3, 4, 5\}$

$(\because (5 + 1)^2 = 36 < 49$ but $(6 + 1)^2 = 49 \nless 49)$

$\therefore$ A and B have the same elements

$\therefore A = B$.

Note :

The number of members in a set

(i) $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

(ii) $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C)$.

Ex. 11 : If $A = \{a, b, c, d, e\}$ and $B = \{d, e, f, g\}$ then obtain $n(A)$, $n(B)$, $n(A \cap B)$ and $n(A \cup B)$.

Hence prove that $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.

Sol. : It is clear that $n(A) = 5$ and $n(B) = 4$.

Now, $A \cup B = \{a, b, c, d, e, f, g\}$ and $A \cap B = \{d, e\}$

$\therefore n(A \cup B) = 7$ and $n(A \cap B) = 2$

$$n(A) + n(B) - n(A \cap B) = 5 + 4 - 2 = 7$$

Thus, $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.

Note : If A and B are disjoint sets then,

$$n(A \cup B) = n(A) + n(B).$$

Ex. 12 : If $A = \{1, 2, 3, 4, 5, 6\}$, $B = \{2, 5, 6, 8\}$ and $C = \{1, 6, 7\}$ then prove that

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C).$$

Sol. : From the given sets it is clear that

$$n(A) = 6, n(B) = 4 \text{ and } n(C) = 3$$

Again, from the given sets

$$A \cup B \cup C = \{1, 2, 3, 4, 5, 6, 7, 8\}$$

$$A \cap B = \{2, 5, 6\}$$

$$B \cap C = \{6\}$$

$$C \cap A = \{1, 6\} \text{ and}$$

$$A \cap B \cap C = \{6\}$$

$$\therefore n(A \cup B \cup C) = 8$$

$$n(A \cap B) = 3, n(B \cap C) = 1,$$

$$n(C \cap A) = 2 \text{ and}$$

$$n(A \cap B \cap C) = 1.$$

From this,

$$\begin{aligned} n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) \\ - n(C \cap A) + n(A \cap B \cap C) \\ = 6 + 4 + 3 - 3 - 1 - 2 + 1 \\ = 8 \end{aligned}$$

$$\begin{aligned} \therefore n(A \cup B \cup C) &= n(A) + n(B) + n(C) - n(A \cap B) \\ &\quad - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C). \end{aligned}$$

Note : If A, B, C are disjoint sets, then

$$n(A \cup B \cup C) = n(A) + n(B) + n(C).$$